

Early-life stress induces persistent astrocyte dysfunction resulting in fear generalisation

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Abstract

Early-life stress can have lifelong consequences, enhancing stress susceptibility and resulting in behavioral and cognitive deficits. While the effects of early-life stress on neuronal function have been well-described, we still know very little about the contribution of non-neuronal brain cells. Investigating the complex interactions between distinct brain cell types is critical to fully understand how cellular changes manifest as behavioral deficits following early-life stress. Here, using male and female mice we report that early-life stress induces anxiety-like behavior and fear generalisation in an amygdala-dependent learning and memory task. These behavioral changes were associated with impaired synaptic plasticity, increased neural excitability, and astrocyte dysfunction. Genetic perturbation of amygdala astrocyte function by either silencing these cells or reducing astrocyte network function was sufficient to replicate cellular, synaptic, and fear memory changes associated with early-life stress. These data provide mechanistic links between early-life stress and astrocyte dysfunction. Our data reveal a role of astrocytes in tuning emotionally salient memory with astrocyte dysfunction resulting in fear generalisation. Further understanding of how astrocytes are affected by stress might offer new insights into the long-term impact of early-life stress on affective states.

eLife assessment

Early-life adversity or stress can enhance stress susceptibility by causing changes in emotion, cognition, and reward-seeking behaviors. This **important** manuscript highlights the involvement of lateral amygdala astrocytes in fear generalization and the associated synaptic plasticity, which are parallel to the effects of early life stress. With an elegant combination of behavioral models, morphological and functional assessments using immunostaining, electrophysiology, and viral-mediated loss-of-function approaches, the authors provide **solid** correlational and causal evidence that is consistent with the hypothesis that early life stress produces neural and behavioral dysfunction via perturbing lateral amygdala astrocytic function.

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Introduction

Both human and rodent studies have provided compelling evidence for the long-lasting effects of adverse early-life experiences, highlighting enhanced susceptibility to subsequent stressors later in life (1–5). Early-life stress (ELS) has widespread effects across the brain and can influence many neural circuits including those involved in threat detection, emotion, cognitive processing, and reward seeking behaviors (6–9). Research investigating the neurobiology of stress has predominantly focused upon investigating the effects on neurons and the consequences for behavior (10–19). We have gleaned tremendous insight into disorders of stress from these studies, yet the links between stress, brain circuits and behavior remain tenuous. Within this framework, the contribution of distinct brain cells, including astrocytes, remains largely untested. Glial cells, which comprise approximately half the cell population in the brain have been shown to directly regulate synaptic transmission and plasticity (20–27), thereby influencing many discrete behaviors (23,28,30–32,34,36). Despite research suggesting important roles for astrocytes in regulating affective states (29), fear (31), and reward seeking behaviors (33,35), whether astrocyte dysfunction prompts stress-induced behavioral impairments remains poorly understood.

First, we characterised whether ELS influenced stress hormone levels across our experimental paradigm from pre-stress until young adulthood. We report that ELS induces a transient increase in blood glucocorticoids that peaks during late adolescence/young adulthood. Turning to behaviour, this increase in glucocorticoid levels was associated with anxiety-like behavior in open field test and elevated plus maze.

Next, we next set out to determine whether ELS induces cognitive impairment using an amygdala-dependent threat discrimination assay. This task was specifically chosen to test the effects of ELS on amygdala function as this brain region is involved with threat detection and associative learning that is acutely sensitive to stress (17,18,37–39). We employed an auditory discriminative fear conditioning paradigm that requires synaptic plasticity changes in the lateral amygdala (40) and found that ELS impairs discriminative fear memory, specifically we found fear generalisation in ELS mice. These hormonal and behavioral changes were associated with impaired synaptic plasticity and enhanced excitability of cells in the lateral amygdala, evidenced by increased c-Fos labelling.

Finally, we investigated the impact of ELS on astrocytes in the lateral amygdala revealing changes in specific proteins associated morphological reorganisation and astrocyte network function. To directly implicate astrocytes with behavioral dysfunction we used two distinct viral approaches to genetically disrupt astrocyte function specifically in the lateral amygdala. These astrocyte-specific manipulations recapitulated ELS-induced fear generalisation, synaptic, and excitability phenotypes supporting the hypothesis that astrocytes play an influential role in the effects of stress on neural circuits and behavior. Together, these findings identify astrocytes as central elements regulating amygdala-dependent affective memory and highlight astrocytes as putative targets for the treatment of stress disorders.

Results

ELS increases blood corticosterone and induces anxiety-like behaviors

We employed an ELS paradigm which combines limited bedding and nesting materials as well as maternal separation from postnatal days 10 to 17 (**Figure 1A** [↗](#)). This ELS paradigm has been shown to induce life-long stress susceptibility ([6](#) [↗](#)). We first set out to determine the impact of this ELS paradigm on stress hormone levels across our experimental window. We took trunk blood from mice across several litters at different time points in the ELS paradigm: pre-stress (P10), end of ELS paradigm (P17), at the start (P45) and end of our experimental window (P70). We report that this ELS paradigm induces a latent increase in blood corticosterone with a significant increase occurring only in late adolescence P45, long after the termination of the stress paradigm (**Figure 1B** [↗](#)).

Next, we measured whether this ELS paradigm, associated with high corticosterone results in long-term changes in anxiety-like behaviors. We found that while there were no changes in behavior in the open-field task (**Figure 1C** [↗](#), **D**), we did see an increase in locomotor activity, i.e. distance travelled, following ELS (**Figure 1E** [↗](#)). Using the elevated plus maze we report increased anxiety-like behavior with a reduction in open-closed arm ration, where a value of 1.0 would indicate equal time spent in each compartment (**Figure 1F** [↗](#), **G**). This was accompanied by an increase in distance travelled within the maze (**Figure 1H** [↗](#)), similar to our findings in the open field. Increased locomotion in response to stress is consistent with previous reports ([15](#) [↗](#)).

Amygdala-dependent memory, plasticity, and excitability are affected by ELS

Considering the strong association between stress disorders, anxiety, and amygdala dysfunction, we next set out to determine whether amygdala-dependent behavior was impacted by ELS. We specifically targeted lateral amygdala-dependent learning and memory using an auditory discriminative fear conditioning paradigm. In brief, during this behavioral task mice are first exposed to a neutral auditory cue (conditioned stimulus -; CS-) and then an aversive cue (CS+) paired with footshock (unconditioned stimulus; US;). On day two mice are placed in a novel environment to remove contextual triggers and exposed to auditory cues from previous day without footshock (**Figure 2A** [↗](#)). Learning and memory were measured using percent time freezing during presentation of auditory cues. We report that mice accurately learn this task and can readily distinguish between neutral and aversive cues (see **Table S1**).

Using this paradigm our data show no difference in freezing responses to the CS- presentations during conditioning between naïve and ELS mice (**Figure 2B** [↗](#), **grey area**). This suggest that ELS does not induce hypersensitivity or hypervigilance to novel auditory cues. We also show that fear acquisition, evidenced by freezing to the CS+/US pairing, is also unaffected by ELS (**Figure 2B** [↗](#)). In contrast, memory recall was affected by stress with ELS mice showing enhanced freezing responses to neutral cues (CS-, **Figure 2C** [↗](#), **grey area**), with no difference in fear responses to aversive cues (CS+; **Figure 2C** [↗](#)) between ELS and naïve mice. Enhanced freezing to neutral auditory cues by ELS mice is indicative of fear generalisation, a common stress-related behavioural phenotype conserved between rodents and humans([41](#) [↗](#), [42](#) [↗](#)). This behavioural observation is more clearly depicted using a discrimination index (DI) that takes into account freezing responses to neutral and aversive cues. Using this measure, DI>0 reflects accurate discrimination, DI=0 no discrimination and DI<0 reflects a discrimination error i.e. higher freezing to the neutral than aversive tone. Using this index, we reveal a significant decrease in fear discrimination in ELS mice compared to naïve mice (**Figure 2D** [↗](#)). This ELS- induced decrease in discrimination can be accounted for by increased freezing to the neutral cue since there are no significant differences in freezing to the aversive cue between ELS and naïve animals. Together these data suggest that ELS enhances affective responses to neutral/non-aversive cues, resulting in fear generalisation in adult mice. Importantly, we observed no differences between male and

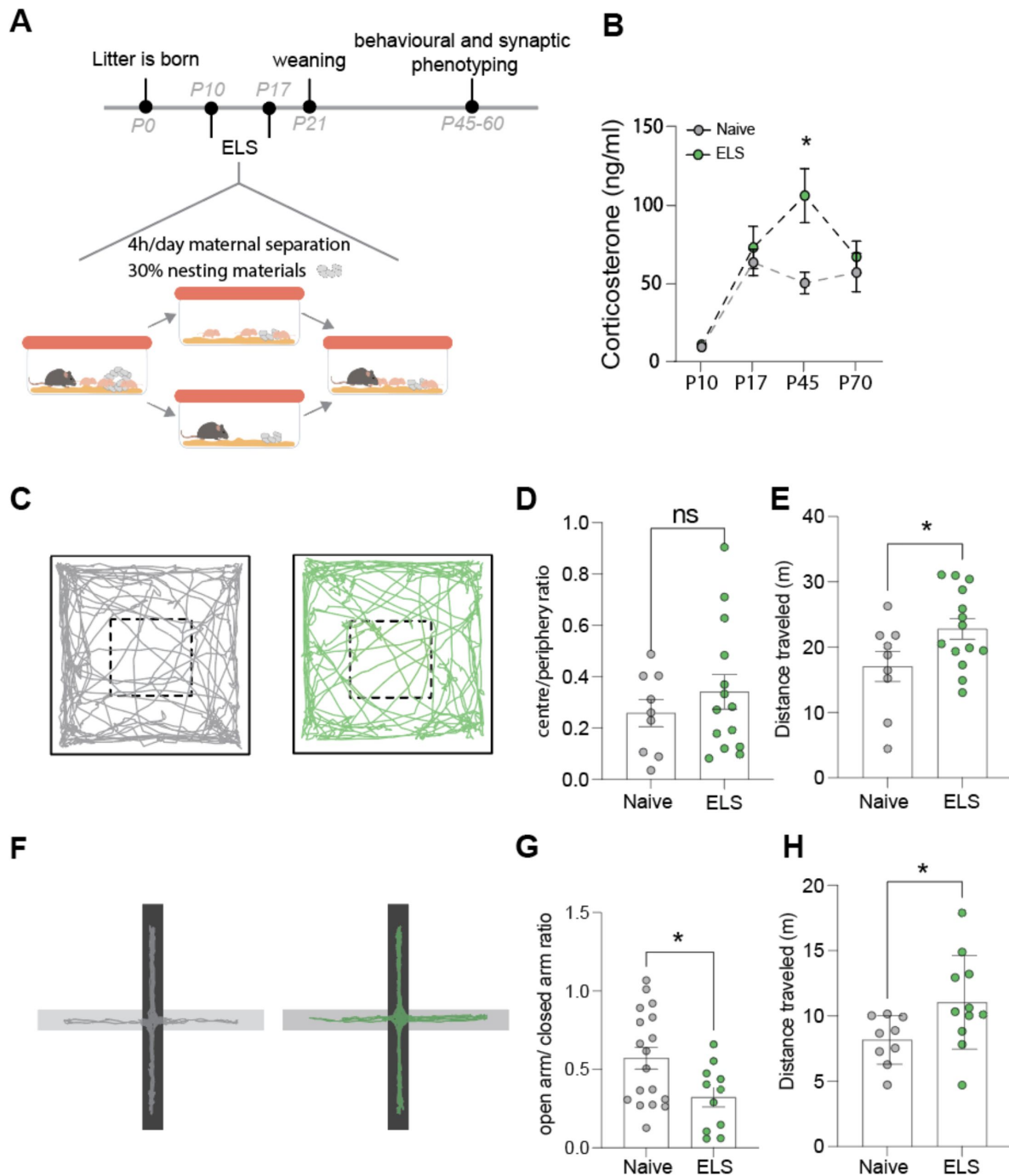


Figure 1.

Effects of ELS on anxiety-like behavior. (A)

Timeline of ELS and behavioral assays. **(B)** Serum corticosterone levels are increased during adolescence, following ELS. $P=0.01$, Naive $N=9$, ELS $N=11$ mice. **(C)** Representative movement tracks of naive (grey) and ELS (green) mice in open field task. **(D)** No significant difference in centre/periphery ratio in open field following ELS. $P=0.39$, Naive $N=9$, ELS $N=14$ mice. **(E)** Total distance travelled during open field test was increased in ELS mice. $P=0.04$, Naive $N=9$, ELS $N=14$ mice. **(F)** Representative locomotor tracks in elevated plus maze in naive and ELS mice, black segment represents closed arms with grey arms representing open arms. **(G)** ELS mice had a reduced open/closed arm ratio. $P=0.02$, Naive $N=18$, ELS $N=11$ mice. **(H)** Total distance travelled during elevated plus maze test was increased by ELS. $P=0.02$, Naive $N=9$, ELS $N=11$ mice. See Table S1 for detailed statistical summaries.

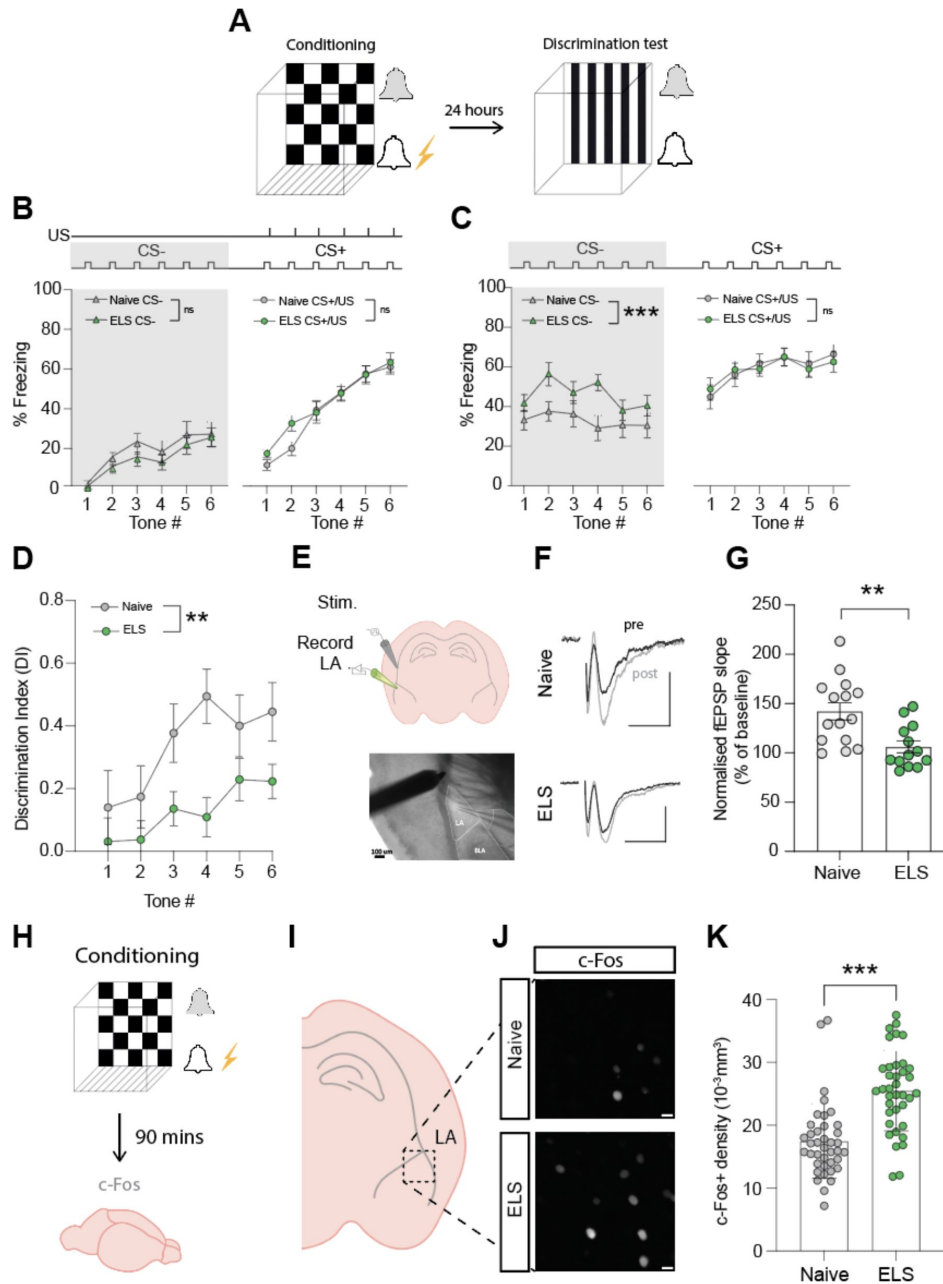


Figure 2.

ELS alters fear memory, synaptic plasticity, and neural excitability. (A)

Timeline of auditory discriminative fear conditioning paradigm. (B) ELS did not affect % Freezing to neutral and aversive auditory cues during learning. Naïve N=20, ELS N=17. (C) % time freezing during presentation of neutral auditory cue (CS-; p=0.0002) and aversive cue (CS+; p=0.99). Naïve N=20, ELS N=17. (D) Discrimination Index was impaired by ELS. P=0.004. Naïve N=20, ELS N=17. (E) Schematic representation of electrode placement in slice electrophysiology experiment (top) with DIC image below. (F) Representative EPSP traces from naïve (top) and ELS (bottom) brain slices during baseline (black trace) and after LTP stimulation (grey trace). Scale bars = 0.5mV and 10ms. (G) LTP was impaired by ELS. P=0.002. Naïve N=15, ELS N=13. (H) Diagram of procedure for c-fos staining experiments. (I) Brain slices containing lateral amygdala (LA) were selected for c-Fos staining. (J) Region of interested extracted from full amygdala image. Scale bars = 10µm. (K) c-Fos positive cell density was increased by ELS. P<0.0001. Naïve n=39 slices N=10 mice, ELS n=37 slices N=10 mice. See Table S1 for detailed statistical summaries.

female mice in this paradigm with both sexes showing equal performance in learning (**Figure S1A**), memory (**Figures S1B**), and auditory discrimination (**Figure S1C**) and use both sexes in all experiments.

Amygdala-dependent auditory fear conditioning has been consistently correlated with the potentiation of lateral amygdala post-synaptic responses to auditory afferents originating from the cortex and the thalamus (35). Click or tap here to enter text. As such, we set out to determine whether ELS impacts synaptic plasticity in cortico-amygdala circuits using acute brain slice electrophysiology and recorded field excitatory postsynaptic potentials (fEPSPs) by stimulating cortical inputs from the external capsule (**Figure 2E**). We found that ELS results in decreased long-term potentiation (LTP) of synaptic transmission in this cortico-amygdala circuit (**Figures 2F,G** and **S1D**). This synaptic observation is consistent with findings using distinct stress paradigms (20), and highlights the important influence of stress on synaptic function across various brain regions.

Fear learning and memory in the LA have also been correlated with the allocation of small subsets of neurons to a fear engram (40). These engrams are thought to be a possible neural substrate for memory (36–37) and their formation depends on neural excitability levels at the moment of acquisition. To reveal stress-induced changes in neural excitability, we collected brain tissue 90 minutes after conditioning paradigm (**Figure 2H**), and carried out immunostaining for the immediate early gene c-Fos, a commonly used activity marker, in the lateral amygdala (**Figures 2I,J**, and **S1E,F**). Fear acquisition in the ELS group resulted in a significant increase in c-Fos density suggestive of increased excitability levels. These results are consistent with other reports of discriminative memory and c-Fos density, with c-Fos density inversely correlated with memory accuracy (45).

Early-life stress induces long-term astrocyte dysfunction

To determine the influence of ELS on astrocytes in the lateral amygdala we carried out immunostaining on brain slices taken from young adult naïve and ELS mice. We first investigated whether increased blood corticosterone affects glucocorticoid receptor (GR) expression and localisation in astrocytes. Using immunostaining we found that this hormone receptor is abundantly expressed in lateral amygdala astrocytes (**Figure 3A**). We further investigated the subcellular localisation of GR within astrocytes. When inactivated GRs are restricted to the cytosol, upon binding to agonists such as corticosterone however, GRs can translocate to the nucleus to influence gene expression (46). Calculating a ratio of cytosolic vs nuclear GR in astrocytes as a proxy for activity, we found an increase in the nuclear fraction of GR in astrocytes following ELS (**Figure 3B**). These results are suggestive of increased activation of glucocorticoid signaling in lateral amygdala astrocytes following ELS.

To assess any potential changes in astrocyte structure and function we investigated expression of the intermediate filament protein glial fibrillary acidic protein (GFAP; **Figure 3C,D**) and found a significant decrease in GFAP fluorescence intensity in the LA following ELS (**Figure 3E**). While functional consequences of GFAP changes are difficult to define, we interpret these modifications in GFAP expression to be indicative of an astrocytic response to stress. To identify specific functional changes induced by ELS, we targeted astrocyte-specific proteins with known roles in influencing synaptic function including Cx43, a GAP-junction protein that composes astrocyte networks (47) and the glutamate transporter GLT-1, responsible for the efficient clearance of glutamate from the synaptic cleft (41; **Figure 3C,D**).

We observed a decrease in fluorescence intensity for Cx43 (**Figure 3F**) while GLT-1 remained unaffected by ELS (**Figure 3G**). These data suggest that astrocytes mount a specific response to ELS which could result in discrete changes in astrocyte network function such as the uncoupling of the astrocytic Cx43 GAP-junction networks in the lateral amygdala.

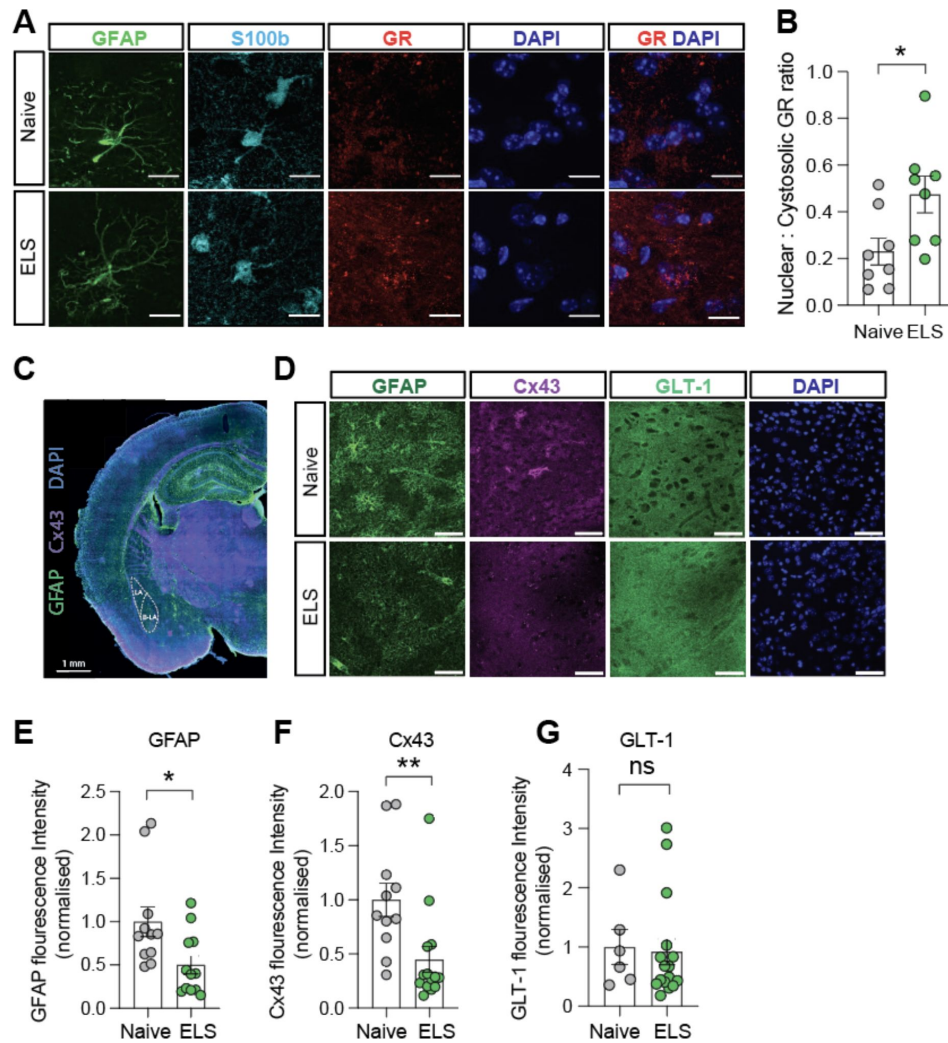


Figure 3.

Long-term astrocyte dysfunction after early-life stress. (A)

Representative immunostaining of GFAP, S100b, GR, DAPI, and GR/DAPI merge in Naïve and ELS conditions. Scale bars = 20µm **(B)** Nuclear/cystolic GR ratio in astrocytes was increased after ELS. $P=0.026$. Naïve $N=8$, ELS $N=8$. **(C)** Representative slide scan image of immunostaining for astrocyte proteins GFAP and Cx43. **(D)** Representative immunostaining of GFAP, Cx43, GLT-1, and DAPI in naïve and ELS conditions. Scale bars = 50µm **(E)** GFAP staining was reduced in ELS mice. $p=0.02$, Naïve $N=11$, ELS $N=12$. **(F)** Cx43 staining was decreased in ELS mice. $p=0.008$. Naïve $N=11$, ELS $N=14$. **(G)** GLT-1 expression was unchanged by ELS. $P=0.8$. Naïve $N=6$, ELS $N=16$. See Table S1 for detailed statistical summaries.

Astrocyte dysfunction alone is sufficient to phenocopy the effects of ELS on fear generalisation, synaptic plasticity, and excitability

To implicate astrocytes in the cellular and behavioral effects of ELS we carried out a series of genetic loss-of-function experiments, specifically targeting astrocytes in the lateral amygdala using viral approaches (**Figure 4A** [↗](#) and **S2A**). Based on the putative astrocyte network dysfunction we identified in fixed tissue (**Figure 3F** [↗](#)), we targeted astrocyte network function by overexpressing a dominant negative Cx43 (dnCx43). We have previously shown this manipulation to completely occlude functional coupling between neighbouring astrocytes ([20](#) [↗](#)). Second, to broadly target astrocyte function we used a calcium extruder pump (hPMCA2/b, a.k.a. CalEx) that has been shown to clamp astrocyte calcium at minimal levels, effectively silencing these cells ([49](#) [↗](#), [50](#) [↗](#)). All viral constructs specifically targeted astrocytes with GfaABC1D promoter and were compared to a control virus expressing eGFP in astrocytes (**Figure 4B** [↗](#)). Initially, we screened for changes in anxiety-like behaviors using elevated plus maze and report no impact of lateral amygdala-specific astrocyte manipulation on the ratio of time spent in the open versus closed arms or in total distance traveled (**Figure S2B, C**). Next, using auditory discriminative fear conditioning paradigm, we found that learning was unaffected by astrocyte dysfunction (**Figure 4C** [↗](#) and **Table S1**). We also report that similar to ELS, lateral amygdala astrocytic dysfunction does not lead to hypersensitivity or hypervigilance to neutral auditory cues (**Figure 4C** [↗](#), **grey area**). During memory recall, however, we found that both astrocyte manipulations enhanced fear responses to neutral cues, without affecting freezing to aversive cues (**Figure 4D** [↗](#)). This behavioural phenotype induced by astrocyte dysfunction alone mimics the effects of ELS on auditory fear discrimination, i.e. enhanced fear generalisation.

Quantification of discrimination index revealed an impairment in discriminative fear memory in both dnCx43 and CalEx conditions, compared to viral control (**Figure 4E** [↗](#)). This effect is dependent on the increase in freezing to the neutral cue with freezing responses to the aversive tone being unaltered by astrocytic dysfunction. These results directly implicate astrocytes in valence processing in the lateral amygdala and suggest a potential link between ELS and astrocyte dysfunction.

Next, we set out to determine whether astrocyte dysfunction in the lateral amygdala induced similar synaptic plasticity deficits in cortico-amygdala circuits as observed in ELS. Again, using acute brain slice electrophysiology, we report an occlusion of LTP in both CalEx and dnCx43 conditions (**Figure 4F** [↗](#)).

Finally, in agreement with fear memory and synaptic plasticity changes, we found that astrocyte dysfunction directly influences neural excitability with increased c-Fos density following fear conditioning in both CalEx and dnCx43 compared to control virus (**Figures 4G,H** [↗](#) and **S2D-F**). Taken together, these data highlight astrocytes as critical regulators of synaptic plasticity, neural excitability, and of discriminative fear memory.

Discussion

In this study, we investigated the potential roles of astrocytes in cellular and behavioral dysfunctions associated with ELS. Using a rodent model of ELS that employs maternal separation and limited bedding, we report increased blood glucocorticoid levels associated with heightened anxiety-like behaviors. These endocrine and behavioural changes were accompanied by impaired discriminative fear memory in an amygdala-dependent auditory discrimination task. More precisely, we observed that ELS mice displayed increased fear responses to neutral cues with no change in freezing responses to aversive stimuli. These behavioral changes were further associated with alterations on the cellular scale including synaptic plasticity and neuronal

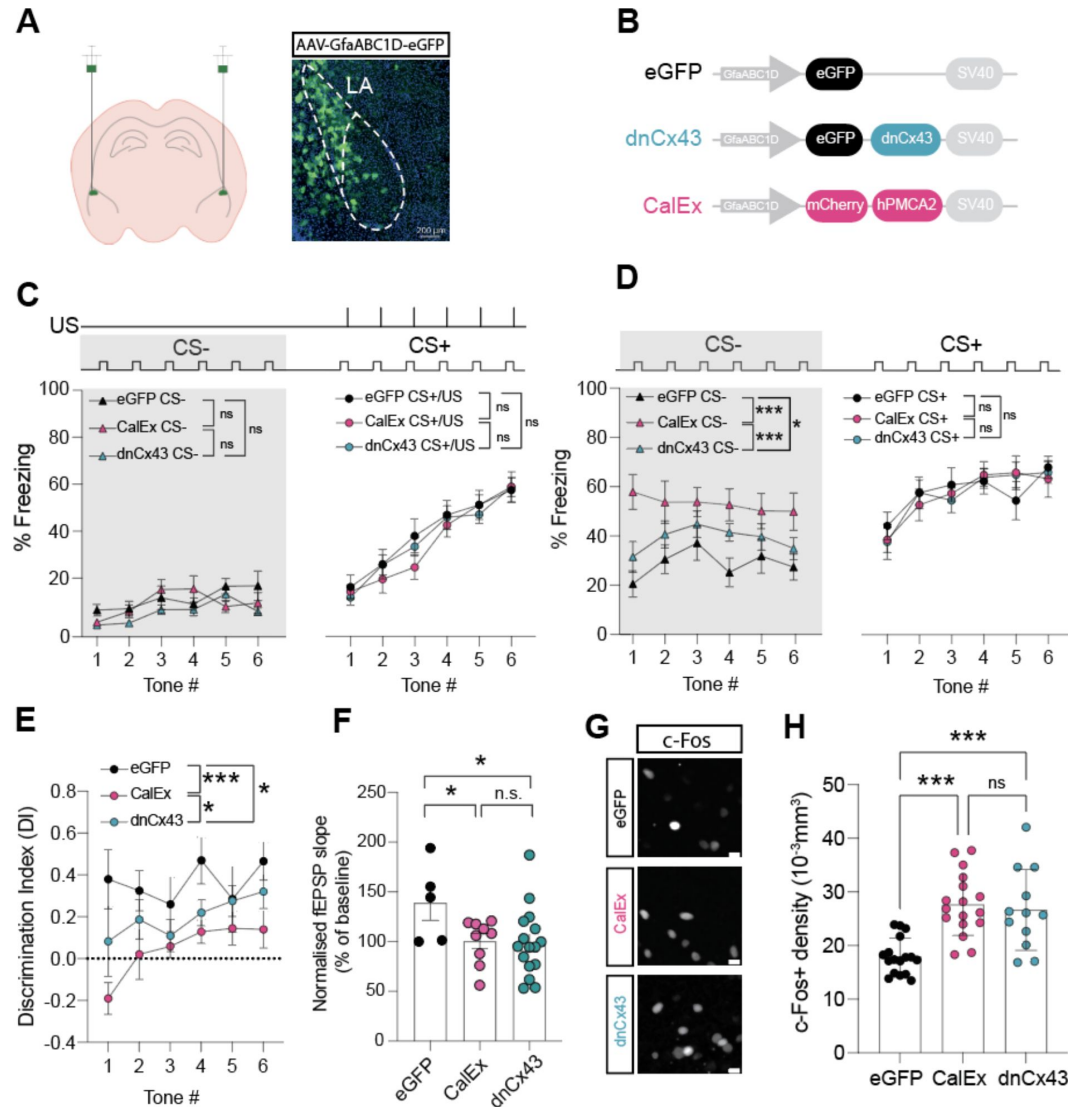


Figure 4.

Astrocyte dysfunction mimics the effects of ELS on behavior, synapses, and neural excitability. (A)

Representative image showing localisation of viral vector in lateral amygdala. **(B)** Viral constructs used to manipulate astrocyte function. **(C)** Astrocyte dysfunction did not affect % Freezing to neutral and aversive auditory cues during learning. eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(D)** % time freezing during presentation of neutral auditory cue (CS-) was increased with astrocyte dysfunction (eGFP vs CalEx: $p < 0.0001$; eGFP vs dnCx43: $p = 0.029$; CalEx vs dnCx43: $p = 0.0005$) with no impact on % Freezing to aversive cue (CS+). eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(E)** Discrimination Indexes were impaired by astrocyte dysfunction. eGFP vs CalEx: $p < 0.0001$; eGFP vs dnCx43: $p = 0.013$; CalEx vs dnCx43: $p = 0.03$. eGFP N=12, CalEx N=11, dnCx43 N=17 mice. **(F)** Astrocyte dysfunction occluded the induction of LTP. eGFP vs CalEx: $p = 0.04$; eGFP vs dnCx43: $p = 0.02$; CalEx vs dnCx43: $p = 0.88$. eGFP N=5, CalEx N=9, dnCx43 N=16. **(G)** Representative c-Fos staining in the lateral amygdala in control (eGFP), CalEx, and dnCx43 conditions. Scale bars = $10 \mu\text{m}$. **(H)** c-Fos positive cell density was increased with astrocyte dysfunction. eGFP vs CalEx: $p < 0.0001$; eGFP vs dnCx43: $p = 0.0005$; CalEx vs dnCx43: $p = 0.88$. eGFP n=17 slices N=5 mice, CalEx n=17 slices N=5 mice, dnCx43 n=12 slices N=5 mice. See Table S1 for detailed statistical summaries.

excitability. Investigating contributions of non-neuronal cells, we found robust changes in lateral amygdala astrocytes following ELS comprising increased glucocorticoid receptor translocation to the nucleus and decreased expression of key astrocyte network proteins. To test the potential role of astrocyte dysfunction and how this might influence stress-induced behavioural changes, we genetically targeted lateral amygdala astrocytes using a viral approach. Astrocyte dysfunction alone, we found, replicated the effects of stress on behavior, synapses, and excitability. Astrocytes have long been shown to be critical for memory formation, with the vast majority of studies reporting that astrocyte dysfunction impairs memory across various brain regions (30 [47](#), 49 [52](#)). Here we show that astrocyte dysfunction in the lateral amygdala effectively impairs discriminative fear memory evidenced by fear generalisation, *i.e.* the inappropriate expression of fear towards stimuli of neutral valence. These data reveal that astrocytes are critical for the appropriate behavioral responses to emotionally salient cues and provide evidence supporting the recently proposed contextual guidance theory of astrocyte function in the brain (51 [52](#)).

Using an ELS paradigm that coincides with peak astrocyte postnatal development (55 [56](#)), we found a latent increase in circulating glucocorticoids after ELS with peak levels appearing during late adolescence and subsiding towards adulthood. This transient effect may contribute to the enhanced susceptibility to stress observed during adolescence (57 [58](#)).

Increased levels of blood glucocorticoids had a direct impact on astrocytes with increased translocation of GR from the cytosol to the nucleus of astrocytes following ELS, indicative of a higher levels of glucocorticoid signaling (46 [47](#)). While the effects of ELS on blood glucocorticoids was transient, this could have long-term consequences on protein expression levels resulting in changes in stress susceptibility. In humans, heightened levels of glucocorticoids in adolescents can be used to predict the occurrence of mood disorders many years later (58 [59](#)).

The behavioural phenotype revealed by this work, *i.e.* ELS- and astrocyte dysfunction-induced fear generalisation, is a common trait observed in anxiety disorders that are commonly associated with fear generalisation (42 [43](#)). As such, these mechanisms give a starting point to further understand the neurobiological changes occurring in the brain in stress disorders.

Our work builds upon a growing literature that is interested in the potential roles of astrocytes in early-life stress (59 [62](#)). Our work identified specific astrocytic impairments induced by ELS and followed up with genetic manipulation of astrocyte function specifically in the lateral amygdala. These manipulations we found replicate cellular and fear generalisation phenotype observed following ELS, providing causal links between astrocyte dysfunction and impaired fear discrimination. We did not, however, observe an anxiety-like phenotype using elevated plus maze and open field, which could be interpreted as a lack of lateral amygdala astrocyte involvement in these discrete behaviors.

We find that both ELS and astrocyte dysfunction both enhance neuronal excitability, assessed by local c-Fos staining in the lateral amygdala following auditory discriminative fear conditioning. One interpretation of these data is that astrocytes might tune engram formation, with astrocyte dysfunction, genetically or after stress, increasing c-Fos expression resulting in a loss of specificity of the memory trace and generalisation of fear. Evidence of a cellular mechanism whereby increased c-Fos levels inversely correlate with memory accuracy have been recently shown in hippocampal circuits (45 [46](#)). In addition, astrocyte-engram neuron interactions have recently been identified (52 [53](#)) suggesting a potential participation of astrocytes in engram stabilisation. Our data are in line with previous reports that have demonstrated clear roles for astrocytes in memory formation and consolidation (31 [53](#), 54 [55](#)) and reveal that astrocytes participate in the neurobiological mechanisms by which stress influences emotional memory.

In sum, our data shed new light on the role of astrocytes as central regulators of amygdala-dependent behavior, synaptic function, and excitability. In addition, our work strongly suggests an astrocytic contribution to stress-induced behavioural and cellular impairments, supporting the notion that astrocytic modifications are not simply downstream of neuronal dysfunction. These ideas are forcing a change in perception of how brain disorders develop, moving away from a neuro-centric cascade and placing astrocytes in the spotlight (51 [↗](#), 64 [↗](#)–68 [↗](#)). Taken together, these data support the hypothesis that astrocytes are key to execute the appropriate behavioral response to emotionally salient cues and highlight astrocytes as potential therapeutic targets for stress-related psychiatric disorders.

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Author Contributions

M.G. I.I.A., L.R.D-H, and C.M-R designed the study. M.G., I.I.A., O.G., J.L-A, J.V. and C.M-R. carried out behavioral experiments, analysing the subsequent data. M.G. and L.R.D-H. carried out immunostaining and analysis of subsequent data. L.R.D-H, M.D., B.R. took blood samples and performed ELISA experiments. M.G., L.R.D-H., O.G., and A.B. carried out injection of viral vectors. M.G. and A.B. carried out electrophysiological experiments and analysis of subsequent data. S.P. maintained and generated mouse colonies. M.G. L.R.D-H, A.B., and C.M-R wrote the manuscript. All authors contributed to ELS paradigm and approved the final version of the manuscript.

Methods

Animals

All experiments were performed in accordance with the guidelines for maintenance and care of animals of the Canadian Council on Animal Care (CCAC) and approved by the Institutional Committee for the Protection of Animals (CIPA) at the Centre Hospitalier de l'Université de Montréal. Both male and female C57BL/6J mice (Jax #000664) were used in the study, housed on a 12hr:12hr light:dark cycle (lights on at 06:30) with *ad libitum* access to food and water.

Early Life Stress protocol

ELS was performed as previously described (6 [↗](#)). C57BL/6J pups were separated from their mothers for 4-hours per day (ZT2-ZT6) for 7 days between ages P10 and P17. During separation pups and mothers were placed into new cages with 60% less bedding. Bedding was weighed on P10 and divided 33% into each cage (home cage, pup separation cage, and mother separation

cage). Separated pups and mothers had *ad libitum* access to food and water and pup cages were placed on a water heating pad maintained at 34 degrees Celsius. After the final day of separation (P17) bedding was returned to the original amount and pups were housed with their mothers until weaning at P21.

Behavioral assays

Auditory Discriminative Fear Conditioning

Mice between ages P45-70 were used. Four days prior to behavioral testing mice were single housed in individual cages. Two days later mice were handled for approximately 10 minutes by hand and mobilisation tube. On the first day of ADFC, single housed mice were left for 1 hour in behavioral room to habituate prior to the start of the conditioning assay.

Conditioning: Mice were first exposed to a neutral stimulus (CS-; white noise, 20s) repeated six times. After a two-minute interval mice were exposed to a tone which becomes the conditioned stimulus (CS+; 12 kHz tone, 20s) as it co-terminates with a mild foot shock (Unconditioned stimulus (US); 0.5mA, 2s) delivered in the last two seconds of tone presentation (repeated six times). Conditioning was carried out in chamber with checkered walls that was cleaned with 70% ethanol between animals.

Memory recall: 24 hours after conditioning mice were exposed to CS- and CS+ (counterbalanced) in a novel environment characterised by stripped walls and a white floor covering, cleaned with 0.5% hydrogen peroxide between animals. Fear learning was quantified by the percentage of time spent freezing to CS- and CS+/US during the conditioning phase. Fear memory was quantified by the percentage time freezing to CS- or CS+ following presentation in the memory-recall phase. A discrimination index was calculated comparing fear responses to CS+ and CS- using the following formula;

$$\text{Discrimination Index (DI)} = \frac{(\text{CS}^+ \text{Freezing}) - (\text{CS}^- \text{Freezing})}{(\text{CS}^+ \text{Freezing}) + (\text{CS}^- \text{Freezing})}$$

Open Field Task

This task comprised of a total of 5 minutes and during that time the mice were left to freely explore the OFT apparatus, a white plexiglass box measuring 38x38 cm. During the task, mice were filmed, and the time spent in distinct parts of the box *i.e.* centre and periphery were quantified using AnyMaze software (Stoelting). The total time spent in the center was used as a quantification of anxiety-like behavior, where more time in the center represented less anxiety-like behavior and the total distance travelled was measured to quantify locomotor changes.

Elevated Plus Maze

This task comprised of a total of 5 minutes during which mice were removed from their cages and placed at the centre of the four arms of the maze, facing an open arm. Mice were video recorded, and these videos were analysed using Anymaze software (Stoelting) which quantified the time spent (s) in closed arms or open arms during the 5min testing period.

Anxiety-like behavior was quantified by ratio of time spent in open vs closed arms.

Acute brain slice preparation

Coronal slices containing lateral amygdala were obtained from mice between P45-70. Animals were anesthetized with isoflurane and the brain was rapidly excised and placed in ice-cold choline-based cutting solution saturated with 95% O₂ and 5% CO₂ containing the following (in mM): 120 choline chloride, 3 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 8 MgCl₂, 20 glucose, pH 7.2 and 295

mOsmol. Slices (300 μ m thick) were cut on a vibratome (Leica VT1200, Nussloch, Germany). Slices were transferred to oxygenated artificial CSF (ACSF) at $32 \pm 0.5^\circ\text{C}$ for 30 min containing the following (in mM): 130 NaCl, 2.5 KCl, 1.25 NaH_2PO_4 , 26 NaHCO_3 , 10 glucose,

1.3 MgCl_2 , 2 CaCl_2 , pH 7.3–7.4 and 300–305 mOsmol, then allowed to recover for at least 30 min before recordings in oxygenated ACSF at room temperature (RT). For experiments, the slices were transferred to a recording chamber where they were perfused (2 ml/min) with ACSF at 32°C for the course of the experiment. Slices were used for a maximum of 6 hr after cutting.

Electrophysiology

Extracellular field excitatory postsynaptic potentials (fEPSPs) were recorded in current clamp mode with a pipette filled with ACSF placed in the lateral part of the amygdala. Monosynaptic EPSPs were elicited by stimulation of cortical afferences with a tungsten concentric bipolar electrode (World Precision Instruments). LTP was induced using a theta burst stimulation (TBS) protocol consisting of 4 pulses at 100 Hz repeated at 5 Hz intervals for 10 seconds. Signals were amplified using a Multiclamp 700B amplifier (Molecular Devices) and digitized using the Digidata 1440 (Molecular Devices). Data were recorded (pClamp 10.2, Molecular Devices) for offline analysis. The magnitude of LTP was quantified by normalizing and averaging the slope of fEPSPs following LTP induction relative to the last 5 min of baseline.

Stereotaxic surgery and AAV delivery

P25–35 mice were given a subcutaneous (s.c.) injection of Carprofen (20 mg/kg) 1h prior surgery. Animals were then deeply anesthetised (2% isoflurane) before placing them into a stereotaxic frame (Kopf instruments). Surgical site was infiltrated with bupivacaine/lidocaine (2mg/kg sc.) 15 min prior surgery. Bilateral injections of the following viral constructs produced at the Canadian Neurophotonics Viral Vector Core: AAV2/5-gfaABC1D-eGFP ($1.2\text{--}1.5 \times 10^{11\text{--}12}$ GC/ml), AAV2/5-gfaABC1D-dnCx43-eGFP ($1.2\text{--}1.7 \times 10^{12}$ GC/ml) and AAV2/5-gfaABC1D-hPMCAw/b-mCherry (1.2×10^{12}) were delivered to the LA (coordinates relative to bregma: -1.4 mm AP, ± 3.4 mm ML and -5 mm DV) of mice of either sex. Viral solutions were injected at a rate of 200 nL/min for a total volume of 650 nL per hemisphere using a 10 μ l Hamilton syringe (Harvard apparatus) and a microinjection syringe pump (UMP3T-2 World Precision Instruments). Animals received 250 μ L of saline for post-surgery hydration.

Brain processing

Mice were anaesthetised in an enclosed chamber filled with isoflurane (5% for induction, 3–4% for maintenance, v/v) until loss of toe pinch reflex. A transcardial perfusion was used to ensure uniform preservation of tissue and brains were fixed in 4 % paraformaldehyde (PFA) for 24 hours at 4 degrees Celsius. The fixed brains were then placed in falcon tubes containing 30% sucrose solution for a minimum of two days. Brains were embedded in optimal cutting temperature (OCT) compound and flash frozen in 2-methylbutane between -40 and -50 degrees Celsius and then stored at -80 until cryosectioning. Brains were cut at 30 μ m thickness at -20 degrees Celsius using a Leica CM3050S Cryostat. Slices were stored in cryoprotector solution at 4 degrees Celsius until immunohistochemistry.

For c-fos labeling experiments, mice were anaesthetised and then perfused 90 minutes after the end of fear conditioning. The whole lateral amygdala was sectioned in 30 μ m thick brain slices and every fourth brain slice was selected for immunohistochemistry.

Immunohistochemistry

Free floating brain slices were washed 3 times in 1X PBS for 15 minutes to remove cryoprotector solution. Slices were permeabilised in block-perm solution (3% Bovine-Serum- Albumin, 0.5% Triton^{10%} in PBS) for a duration of 1 hour. Slices were incubated with antibodies for 24 hours at 4

degrees Celsius. Slices were incubated with primary antibodies; [Rabbit] anti-S100 β (1:1000, Abcam, ab52642), [Mouse] anti-glucocorticoid receptor (1:500, ThermoFisher, MA1-510), [Chicken] anti-GFAP (1:1000, ThermoFisher, PA1-10004), [mouse] anti-Cx43 (1:1000 ThermoFisher, 3D8A5), [Rabbit] Anti-GLT-1 (EAAT2) (1:1000, NovusBio, NBP1- 20136) and), [Rabbit] Anti-c-Fos (1:2000, Synaptic Systems, Cat.No. 226008). After primary antibody incubation slices were washed 3 times in 1X PBS for 15 minutes to remove non-specific binding. Slices were incubated with secondary antibodies in a DAPI solution (1:10000) at RT in aluminium foil for 1 hour. Slices were incubated with secondary antibodies; [goat] anti-rabbit Alexa 488 (1:1000, Jackson Immuno Research, 111-545-144), [goat] anti-mouse Alexa 647 (1:1000, ThermoFisher, A32728), [goat] anti-chicken Alexa 568 (1:1000, ThermoFisher, A11041). After secondary antibody incubation slices were washed again 3 times in 1X PBS for 15 minutes before being mounted onto Fisherbrand microscope slides using ProLong™ Glass Antifade mountant (P36982).

Microscopy

Slices were imaged using a Leica TCS SP5 laser scanning confocal microscope with oil immersion Plan-Apochromat 63x objective 1.4 NA, or using a Zeiss Observer Z1 spinning disk confocal microscope/TIRF with a 20x objective 1.8 NA. 16-bit images of 246 x 246 μm areas were acquired at 400Hz (frame size (x*y); 1024 x 1024, pixel size; 250nm). 25-30 μm z-stacks were acquired with a step size of 0.5 μm . 15 μm max intensity z-projections of the lateral amygdala were analysed using Image-J (Fiji) to obtain A.U fluorescence intensity measures of secondary antibodies attached to primaries with specificity to epitopes; GFAP, Cx43, GLT-1, and GR. We applied a fluorescence threshold for GFAP, Cx43 and GLT-1 fluorescence and measured integrated density A.U (thresholds; Mean, Otsu, Otsu respectively). For nuclear and cytosolic Glucocorticoid Receptor measures, astrocyte (s100b⁺) DAPI nuclei were thresholded and used as ROIs for nuclear measures of GR fluorescence (S100b⁺ + DAPI⁺). The same nuclear DAPI ROI was cleared from the previously thresholded s100b regions to obtain cytosolic astrocytic GR (S100b⁺ - DAPI⁺). Integrated density A.U of GR fluorescence was used to calculate Nuclear:Cytosolic GR ratios.

For c-Fos cell counting in the lateral amygdala, 16-bit images containing the lateral and basal amygdala nuclei were acquired. 20-30 z-stacks were acquired with a step size of 1 μm . For c-Fos particle counting analysis, a region of interest was used to isolate the lateral amygdala and then 8-bit z-stacks of the c-Fos channel were analysed semi-automatically by using the FIJI plugin Quantify-cfos (69 [DOI](#)). In brief, a threshold for area and fluorescence intensity was set manually to identify ROIs around all c-Fos positive nuclei. These ROIs were then manually inspected to remove false positives and negatives found by the plug-in. c-Fos particle counts were then normalized by the volume of the z-stack to calculate the c-Fos density. In order to calculate the volume, the area of the lateral amygdala was multiplied by the number of z-stacks.

Blood Collection & Corticosterone analysis

Trunk blood was collected via decapitation at ZT2 (08:30). To minimise handling induced elevations in corticosterone mice were housed with clear plastic tubes used to move individual mice into an enclosed chamber filled with 5% isoflurane for 2 minutes. Mice were placed in enclosed Isoflurane until loss of toe pinch reflect (maximum elapsed time of 2 minutes). Mice were decapitated and trunk blood was collected into BD 365963 Microtainer® Capillary Blood Collection Tubes and placed onto Ice. Blood was centrifuged for 5 minutes at 4 degrees Celsius at 5000rpm. Serum was aliquoted and stored at -80 degrees Celsius. Corticosterone measurements were obtained using an ENZO ELISA kit (ADI-900-097).

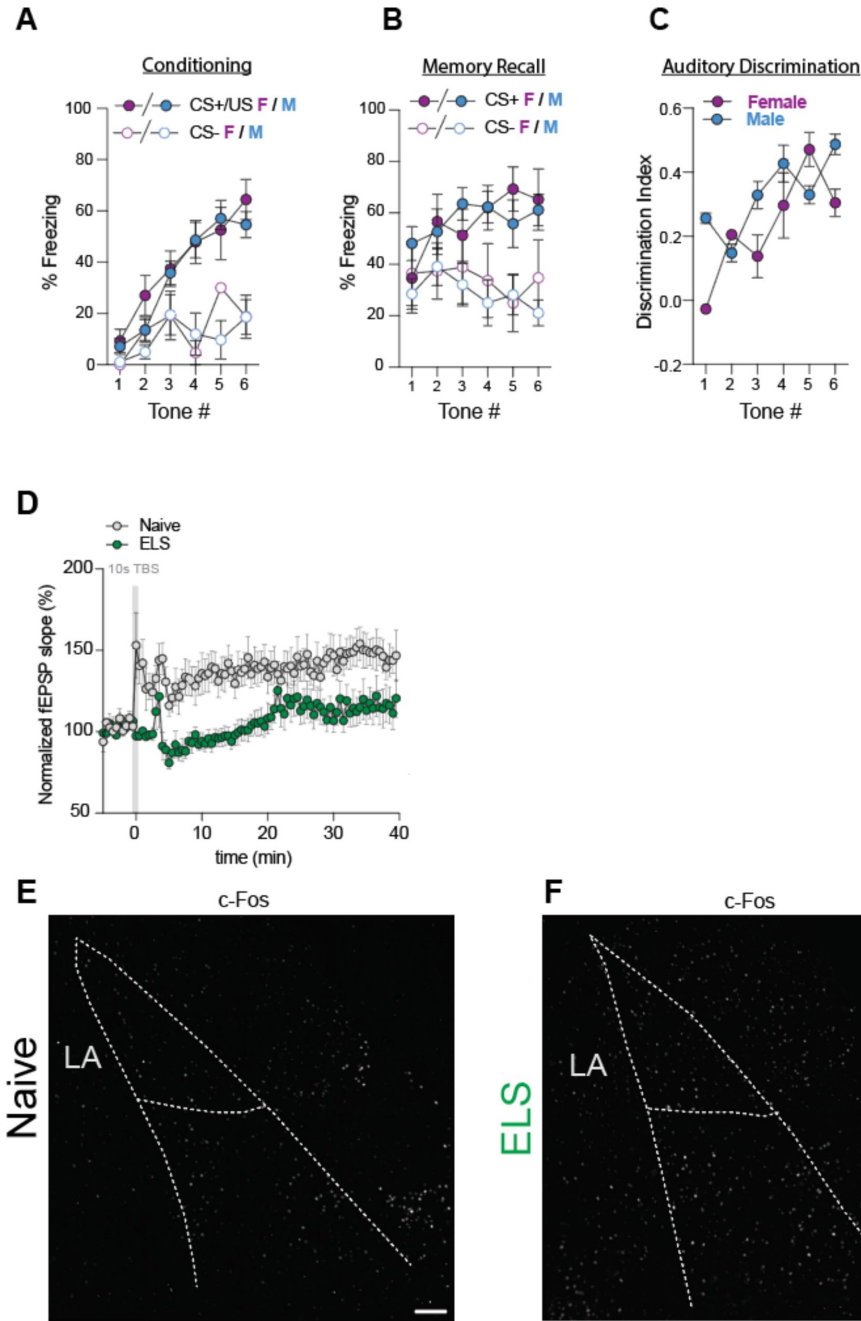
Statistical analyses

Results are presented as mean $\pm$ S.E.M. Data with one variable were analyzed with the two-tailed Student's *t* test or Mann-Whitney test. Data with more than two conditions were first screened for a Gaussian distribution with Kolmogorov-Smirnov test followed by analysis either with one-

way/repeated measures ANOVA or Kruskal-Wallis/Friedman test when needed and Tukey's multiple-comparison parametric *post hoc* test (data with Gaussian distribution) or by a Dunn's multiple-comparison non-parametric *post hoc* test (data with non-Gaussian distribution). Auditory discriminative fear conditioning data was analysed using a two-way ANOVA with appropriate post-hoc tests. Open field and elevated plus maze data analysed using one-way ANOVA with appropriate post hoc tests. Graphic significance levels were *, $p < 0.05$; **, $p < 0.01$ and ***, $p < 0.001$. All data were analyzed using GraphPad Prism software (Version 9, GraphPad, USA).

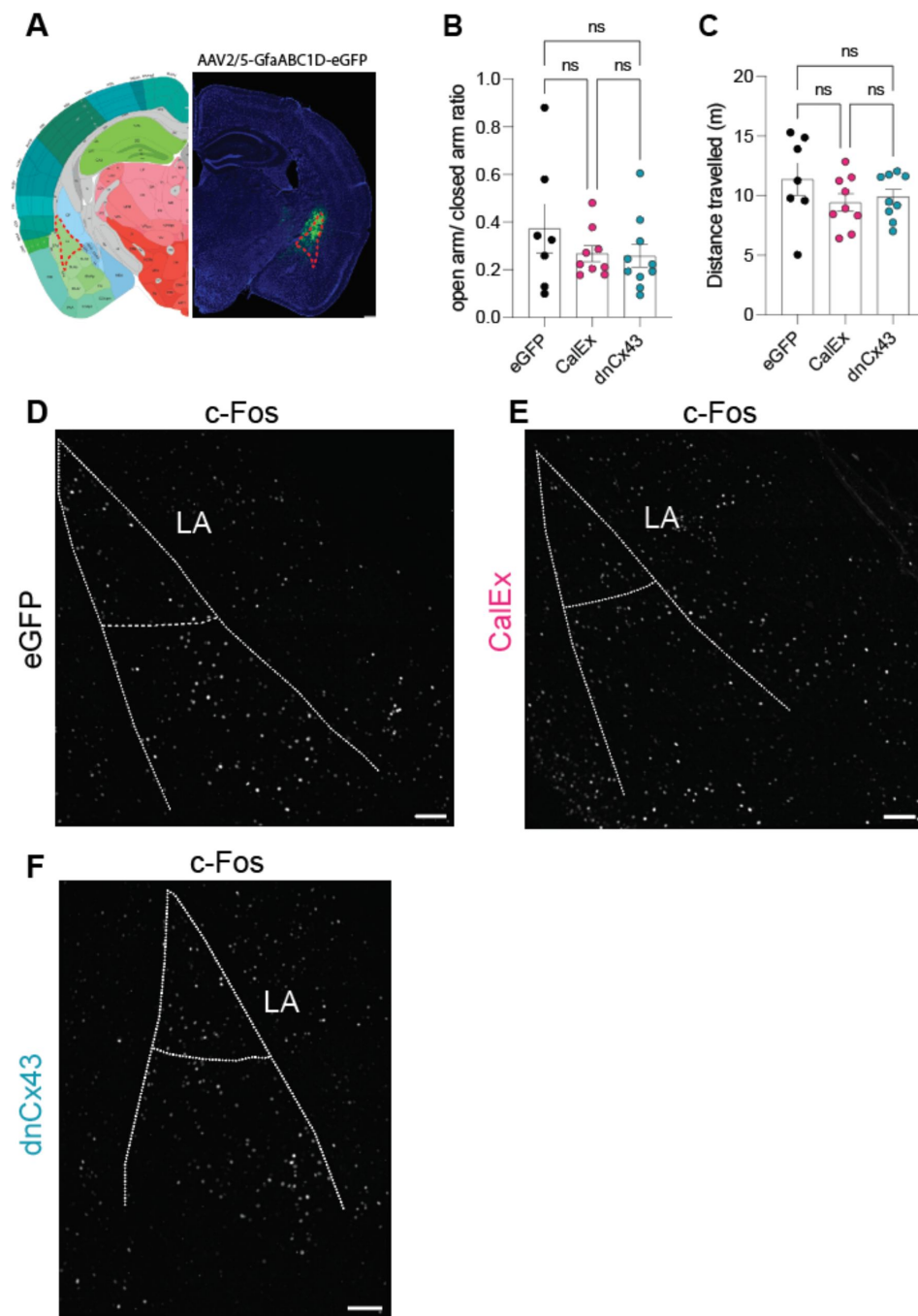
Data availability

The datasets that support the findings of this study are available from the corresponding author upon request.



Supplementary Figure 1

(A) Learning phase (conditioning) of auditory discriminative fear conditioning in male and female naïve mice. Female N=7, Male N=8. (B) Memory recall in of CS+ and CS- in male and female naïve mice. Female N=7, Male N=8. (C) Auditory discrimination in male and female naïve mice. Female N=7, Male N=8. (D) Normalised fEPSP slope showing synaptic response before and after theta burst stimulation (grey vertical bar) in naïve and ELS mice. (E) Representative Z-stack projection of c-Fos staining in entire LA and BLA in naïve. (F) Z-stack projection of c-Fos staining in entire LA and BLA in ELS. Scale bars = 100µm. See Table S1 for detailed statistical comparisons.



Supplementary figure 2

(A) Representative image of fluorescent reporter expression in lateral amygdala following AAV injection. **(B)** No effect of astrocyte manipulation on open arm/closed arm duration in EPM. eGFP N=7, CalEx N=9, dnCx43 N=9 mice. **(C)** No effect of astrocyte manipulation on distance travelled in EPM. eGFP N=7, CalEx N=9, dnCx43 N=9 mice. **(D)** Representative Z-stack projection of c-Fos staining in entire LA and BLA in eGFP injected mouse 90minutes after conditioning. **(E)** Representative Z-stack projection of c-Fos staining in entire LA and BLA in CalEx injected mouse 90minutes after conditioning. **(F)** Representative Z-stack projection of c-Fos staining in entire LA and BLA in dnCx43 injected mouse 90minutes after conditioning. Scale bars = 100µm

Figure 1	
1B	CORT at P45 P=0.01; Naïve=50.1±6.9 ng/ml, N=9; ELS=106.1±17.2 ng/ml, N=11; mean±s.e.m
1D	Centre/periphery ratio Naïve vs ELS unpaired t-test, P=0.3916; Naïve=0.2585±0.05273 a.u, N=9; ELS=0.3414±0.06774 a.u, N=14; mean±s.e.m
1E	Total distance travelled Naïve vs ELS unpaired t-test, P=0.0454; Naïve= 17.04± 2.300 m, N=9; ELS= 22.82± 1.602 m, N=14; mean±s.e.m
1G	Open arm/closed arm ratio Naïve vs ELS unpaired t-test, P=0.0224; Naïve= 0.5721± 0.07017 a.u, N=18 ELS= 0.3237± 0.06254 a.u, N=11; mean±s.e.m
1H	Total distance travelled unpaired t-test, P=0.0224; Naïve= 8.167± 0.6223 m, N=9; ELS= 11.03± 1.079 m, N=11; mean±s.e.m
Figure 2	
2B	Fear conditioning CS- CS+ Naïve vs ELS Two-way ANOVA; effect of group: F (3, 393) = 64.85, P<0.0001; Tukey's multiple comparison test, Naïve CS- vs ELS CS-: P=0.3602, Naïve CS+ vs ELS CS+: P=0.5618, Naïve CS- vs Naïve CS+: P<0.0001, ELS CS- vs ELS CS+: P<0.0001, Naïve CS-=17.92±3.734 % Freezing, N=20 mice Naïve CS+=38.55±8.204 % Freezing, N=20 mice ELS CS-=13.55±3.491 % Freezing, N=17 mice ELS CS+=41.67±6.869 %Freezing, N=17 mice Mean ± s.e.m
2C	Discrimination test Naïve vs ELS Two-way ANOVA; effect of group: F (3, 435) = 32.36, P<0.0001; Tukey's multiple comparison test, Naïve CS- vs ELS CS-: P=0.0002, Naïve CS+ vs ELS CS+: P=0.9991, Naïve CS-=32.9±1.4 % Freezing., N=20 mice

Supplementary Table 1

	ELS CS- = 46.03 ± 2.9 % Freezing., N=17 mice Naïve CS+ = 58.77 ± 3.2 % Freezing., N=20 mice ELS CS+ = 58.34 ± 2.2 % Freezing., N=17 mice Mean ± s.e.m
2D	Discrimination index Naïve vs ELS Two-way ANOVA; effect of group: $F(1, 35) = 9.457$, $P = 0.0041$; Naïve = 0.3384 ± 0.05986; N=20 mice ELS = 0.1275 ± 0.03533, N=17 mice mean ± s.e.m
2G	LTP Naïve vs ELS Unpaired t-test, $P = 0.0028$; Naïve = 142.1 ± 8.720%, N=15; ELS = 106.0 ± 6.058, N=13; mean ± s.e.m
2K	c-Fos density Naïve vs ELS Mann-Whitney test, $P < 0.0001$, $U = 228$, Naïve = 17.42 ± 0.9427 c-Fos density/10⁻³mm³, n=39 slices from N=10 mice; ELS = 25.45 ± 1.044 c-Fos density/10⁻³mm³, n=37 slices from N=10 mice; mean ± s.e.m.
Figure 3	
3B	Nuclear:cytosolic GR ratio Naïve vs ELS $P = 0.026$; Naïve = 0.23 ± 0.06 a.u., N=8; ELS = 0.48 ± 0.08 a.u., N=8; mean ± s.e.m.
3E	GFAP $P = 0.02$, unpaired t-test; Naïve = 1 ± 0.17 a.u., N=11; ELS = 0.5 ± 0.1 a.u, N=12; mean ± s.e.m.
3F	Cx43 $P = 0.008$, unpaired t-test; Naïve = 1 ± 0.15 a.u., N=11; ELS = 0.45 ± 0.12 a.u., N=14; mean ± s.e.m.
3G	GLT-1 $P = 0.8$, unpaired t-test; naïve = 1 ± 0.29 a.u., N=6; ELS = 0.9 ± 0.21 a.u., N=16; mean ± s.e.m
Figure 4	
4C	Fear conditioning CS- CS+ eGFP vs CalEx vs dnCx43 Two-way ANOVA; effect of group: $F(5, 463) = 78.39$, $P < 0.0001$; Tukey's multiple comparison test, eGFP CS- vs dnCx43 CS-: $P = 0.1940$, eGFP CS- vs CalEx CS-: $P = 0.9758$, dnCx43 CS- vs CalEx CS-: $P = 0.7027$, eGFP CS+ vs dnCx43 CS+: $P = 0.9536$, eGFP CS+ vs CalEx CS+: $P = 0.6680$,

Supplementary Table 1 (continued)

	dnCx43 CS+ vs CalEx CS+: $P=0.9761$, eGFP CS-=11.24 ± 1.875 % Freezing., N=12 mice CalEx CS-=9.182 ± 2.247 % Freezing., N=11 mice dnCx43 CS-=5.515 ± 1.936 % Freezing., N=17 mice eGFP CS+= 39.33 ± 6.443 % Freezing., N=12 mice CalEx CS+=35.22 ± 7.464 % Freezing., N=11 mice dnCx43 CS+=37.14 ± 6.923 % Freezing., N=17 mice mean$\pm$ s.e.m
4D	Discrimination test eGFP vs CalEx vs dnCx43 freezing responses Two-way ANOVA; effect of group: $F(5, 457) = 25.29$, $P<0.0001$; Tukey's multiple comparison test, eGFP CS- vs dnCx43 CS-: $P=0.0293$, eGFP CS- vs CalEx CS-: $P<0.0001$, dnCx43 vs CalEx: $P=0.0005$, eGFP CS+ vs dnCx43 CS+: $P>0.9999$, eGFP CS+ vs CalEx CS+: $P>0.9999$, dnCx43 CS+ vs CalEx CS+: $P>0.9999$ eGFP CS-= 28.73 ± 2.342 % Freezing., N=12 mice CalEx CS-= 53.00 ± 1.203 % freezing., N=11 mice dnCx43 CS-= 38.81 ± 1.957 % Freezing., N=17 mice eGFP CS+= 57.79 ± 3.314 % Freezing., N=12 mice CalEx CS+= 57.02 ± 4.219 % Freezing., N=11 mice dnCx43 CS+= 57.32 ± 4.344 % Freezing., N=17 mice mean$\pm$ s.e.m.
4F	LTP eGFP vs CalEx vs dnCx43 Ordinary one-way ANOVA, group effect: $F(2, 31) = 4.221$, $P=0.0239$; Uncorrected Fisher's LSD, eGFP vs CalEx: $P=0.0408$, eGFP vs dnCx43: $P=0.0207$, CalEx vs dnCx43: $P=0.8848$, eGFP=139.0 ± 17.70 % baseline., N=5; CalEx=100.3 ± 7.485 % baseline., N=9; dnCx43=98.31 ± 8.635 % baseline., N=16; mean$\pm$ s.e.m
4H	c-Fos density eGFP vs CalEx vs dnCx43 Ordinary one-way ANOVA, group effect: $F(2, 43) = 14.91$, $P<0.0001$; Tukey's multiple comparison test, eGFP vs CalEx: $P<0.0001$, eGFP vs dnCx43: $P=0.0005$, CalEx vs dnCx43: $P=0.8807$

Supplementary Table 1 (continued)

	eGFP=17.97±0.8195 c-Fos density/10 ⁻³ mm ³ , n=17 slices from N=5 mice; CalEx=27.66±1.398 c-Fos density/10 ⁻³ mm ³ , n=17 from N=5 mice; dnCx43=26.65±2.173 c-Fos density/10 ⁻³ mm ³ , n=12 slices from N=5 mice; mean±s.e.m
Supplementary figures	
S1A	Fear conditioning CS- CS+ Females vs Males Two-way ANOVA; effect of group: F (3, 99) = 15.54, P<0.000; Tukey's multiple comparison test, Female CS- vs Male CS-: P=0.2521, Female CS+ vs Male CS+: P=0.8432, Female CS- vs Female CS+: P=0.0017, Male CS- vs Male CS+: P<0.0001, Female CS-=23.90±5.126 % Freezing., N=7 mice Male CS-=16.31±3.339 % Freezing., N=8 mice, Female CS+=39.49±8.089 % Freezing., N=7 mice, Male CS+=36.1±8.544 % Freezing., N=8 mice, Mean±s.e.m.
S1B	Discrimination test Females vs Males Two-way ANOVA; effect of group: F (3, 156) = 13.92, P<0.0001; Tukey's multiple comparison test, Female CS- vs Male CS-: P=0.3956, Female CS+ vs Male CS+: P=0.8851, Female CS- vs Female CS+: P=0.0011, Male CS- vs Male CS+: P<0.0001, Female CS-=36.79±1.241 % Freezing., N=7 mice, Male CS-=27.97±2.183 % Freezing., N=8 mice, Female CS+=58.83±4.849 % Freezing., N=7 mice, Male CS+=54.75±2.580 % Freezing., N=8 mice, Mean±s.e.m.
S1C	Discrimination index Females vs Males Two-way ANOVA; effect of group: F (1, 13) = 0.03698, P=0.8505; Female DI=0.3320±0.06280 a.u., N=7 mice, Male DI=0.3627±0.06860 a.u., N=8 mice, Mean±s.e.m.
S2B	Open arm/Closed arm ratio eGFP vs CalEx vs dnCx43 Two-way ANOVA; effect of group: F (2, 23) = 0.9903, P=0.3867; Tukey's multiple comparison test, eGFP vs CalEx: P=0.4961, eGFP vs dnCx43: P=0.3919, CalEx vs dnCx43: P=0.9783,

Supplementary Table 1 (continued)

	eGFP=0.3739±0.1036 a.u., N=7 mice, CalEx=0.2679±0.03393 a.u., N=9 mice, dnCx43=0.2576±0.05335a.u., N=9 mice, Mean±s.e.m.
S2C	Distance traveled eGFP vs CalEx vs dnCx43 Ordinary one-way ANOVA; effect of group: F (2, 24) = 1.304, P=0.2901; Tukey's multiple comparison test, eGFP vs CalEx: P=0.2722, eGFP vs dnCx43: P=0.4726, CalEx vs dnCx43: P=0.9265, eGFP=11.38±1.379 m., N=7 mice, CalEx=9.482±0.5968 m., N=9 mice, dnCx43=9.898±0.6352 m., N=9 mice, Mean±s.e.m.

Supplementary Table 1 (continued)

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Reviewer #1 (Public Review):

Summary:

The manuscript asks the question of whether astrocytes contribute to behavioral deficits triggered by early life stress. This question is tested by experiments that monitor the effects of early life stress on anxiety-like behaviors, long-term potentiation in the lateral amygdala, and immunohistochemistry of astrocyte-specific (GFAP, Cx43, GLT-1) and general activity (c-Fos)

markers. Secondly, astrocyte activity in the lateral amygdala is impaired by viruses that suppress gap-junction coupling or reduce astrocyte Ca^{2+} followed by behavioral, synaptic plasticity, and c-Fos staining. Early life stress is found to reduce the expression of GFAP and Cx43 and to induce translocation of the glucocorticoid receptor to astrocytic nuclei. Both early life stress and astrocyte manipulations are found to result in the generalization of fear to neutral auditory cues. All of the experiments are done well with appropriate statistics and control groups. The manuscript is very well-written and the data are presented clearly. The authors' conclusion that lateral amygdala astrocytes regulate amygdala-dependent behaviors is strongly supported by the data. However, the extent to which astrocytes contribute to behavioral and neuronal consequences of early life stress remains open to debate.

Strengths:

A strong combination of behavioral, electrophysiology, and immunostaining approaches is utilized and possible sex differences in behavioral data are considered. The experiments clearly demonstrate that disruption of astrocyte networks or reduction of astrocyte Ca^{2+} provokes generalization of fear and impairs long-term potentiation in the lateral amygdala. The provocative finding that astrocyte dysfunction accounts for a subset of behavioral effects of early life stress (e.g. not elevated plus or distance traveled observations) is also perceived as a strength.

Weaknesses:

The main weakness is the absence of more direct evidence that behavioral and neuronal plasticity after early life stress can be attributed to astrocytes. It remains unknown what would happen if astrocyte activity were disrupted concurrently with early life stress or if the facilitation of astrocyte Ca^{2+} would attenuate early life stress outcomes. As is, the only evidence that early life stress involves astrocytes is nuclear translocation of GR and downregulation of GFAP and Cx43 in Figure 3 which may or may not provoke astrocyte Ca^{2+} or astrocyte network activity changes.

<https://doi.org/10.7554/eLife.99988.1.sa2>

Reviewer #2 (Public Review):

Summary:

In this manuscript, Guayasamin et al. show that early-life stress (ELS) can induce a shift in fear generalisation in mice. They took advantage of a fear conditioning paradigm followed by a discrimination test and complemented learning and memory findings with measurements for anxiety-like behaviors. Next, astrocytic dysfunction in the lateral amygdala was investigated at the cellular level by combining staining for c-Fos with astrocyte-related proteins. Changes in excitatory neurotransmission were observed in acute brain slices after ELS suggesting impaired communication between neurons and astrocytes. To confirm the causality of astrocytic-neuronal dysfunction in behavioral changes, viral manipulations were performed in unstressed mice. Occlusion of functional coupling with a dominant negative construct for gap junction connexin 43 or reduction in astrocytic calcium with CalEx mimicked the behavioral changes observed after ELS suggesting that dysfunction of the astrocytic network underlies ELS-induced memory impairments.

Strengths:

Overall, this well-written manuscript highlights a key role for astrocytes in regulating stress-induced behavioral and synaptic deficits in the lateral amygdala in the context of ELS. Results are innovative, and methodological approaches relevant to decipher the role of astrocytes in

behaviors. As mentioned by the authors, non-neuronal cells are receiving increasing attention in the neuroscience, stress, and psychiatry fields.

Weaknesses:

I do have several suggestions and comments to address that I believe will improve the clarity and impact of the work. For example, there is currently a lack of information on the timeline for behavioral experiments, tissue collection, etc.

<https://doi.org/10.7554/eLife.99988.1.sa1>

Reviewer #3 (Public Review):

Summary

The authors show that ELS induces a number of brain and behavioral changes in the adult lateral amygdala. These changes include enduring astrocytic dysfunction, and inducing astrocytic dysfunction via genetic interventions is sufficient to phenocopy the behavioral and neural phenotypes. This suggests that astrocyte dysfunction may play a causal role in ELS-associated pathologies.

Strengths:

A strength is the shift in focus to astrocytes to understand how ELS alters adult behavior.

Weaknesses:

The mechanistic links between some of the correlates - altered astrocytic function, changes in neural excitability, and synaptic plasticity in the lateral amygdala and behaviour - are underdeveloped.

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